WildGuard

Cross-dataset label transfer for wildlife-trade risk prioritisation

GROUP 37 | PRS PRACTICE MODULE | FINAL PRESENTATION

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The problem: more trade than anyone can inspect

Scale beyond manual review

CITES full download v2026.1: 29.2 million records since 1975; ~825 thousand animal records a year in 2015-2024.

The signal is a pattern, not a field

Risk shows up in combinations - species x product x quantity x purpose x source x route - judged against what is normal for that species.

A recognised need, not yet tooled

CITES Secretariat & WCO guidance on risk assessment and inspections for CITES trade controls (CoP20 Decisions 20.130-20.131, Dec 2025): targeting is still largely manual.

How we got here: from image fusion to label transfer

Original proposal: fuse CITES records with pangolin images into one review-priority score

Modality & fairness

Images covered pangolins only; records covered every species - missing images would skew scores either way.

Usefulness

Would mainly repeat known species-level risk, not say which specific shipment is urgent.

Data feasibility

Image dataset request unanswered; synthetic augmentation brings a synthetic-to-real gap.

Decision: drop the visual + structured fusion design

New direction: LEMIS + CITES, one modality, cross-dataset label transfer

The idea: LEMIS has the labels, CITES has the scale

LEMIS · US Fish & Wildlife Service

gives LABELS

- 5.51M US import line items, 2000-2014
- Real outcomes: refused / seized / abandoned / re-exported vs cleared
- Enforcement cases: 1.74% of line items - heavily imbalanced
- Lacks: global coverage

schema alignment + label transfer

CITES Trade Database

gives SCALE

- 8.25M animal trade records scored, 2015-2024
- Every Party, every CITES-listed animal taxon
- No enforcement labels (illegal-trade DB is restricted)
- Lacks: labels, declared value

LEMIS v1.1.0 (Zenodo, CC-BY-4.0) · CITES full download v2026.1 (UNEP-WCMC for the CITES Secretariat)

Making two datasets speak the same language

Field	LEMIS	CITES	Shared form
Product form	94 codes (LIV, SKI, IVC …)	101 terms (live, skins …)	31 product groups
Unit	NO, KG, GM, ML …	kg, g, blank = number …	count / kg / litre / m / m^2 / m^3
Source, purpose	CITES letter codes	CITES letter codes	identical
Country	ISO-2	ISO-2	identical + world region
CITES appendix	not recorded	recorded	looked up for LEMIS taxa
Declared value	yes (USD)	no	LEMIS-only model

Alignment check

Can a classifier tell the datasets apart? (AUC 0.5 = indistinguishable)

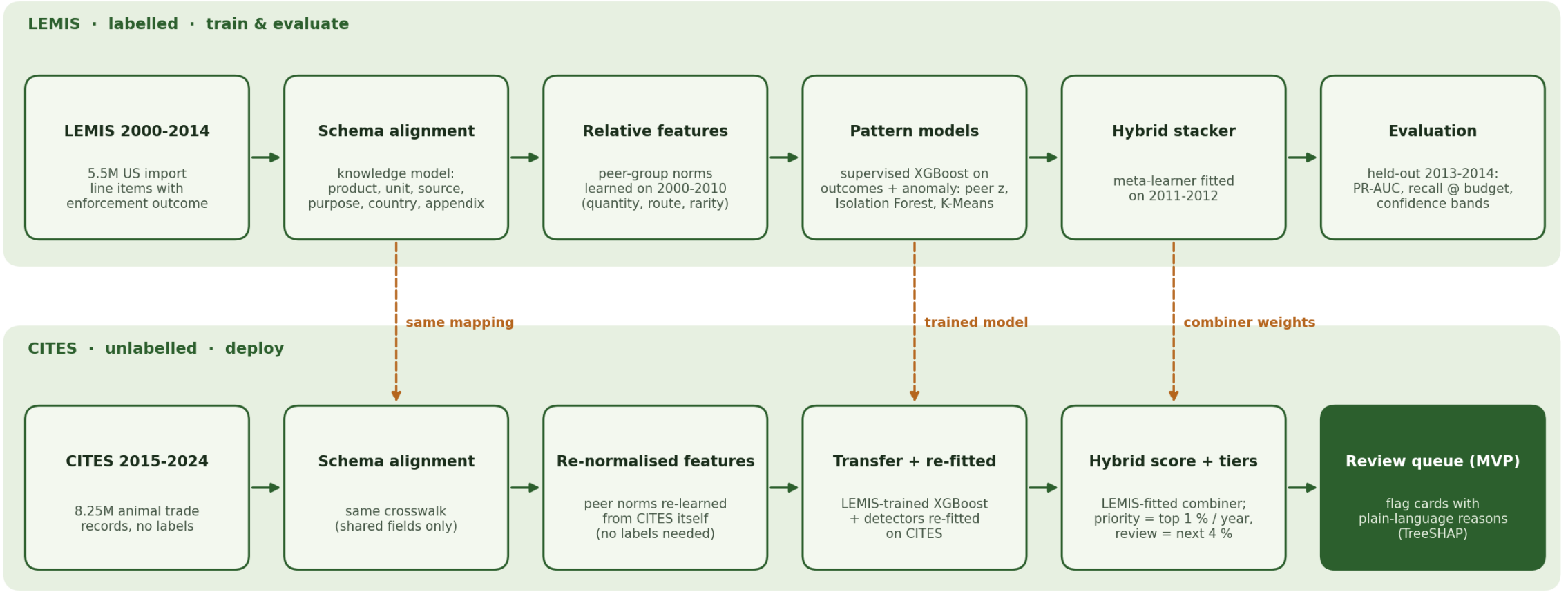
0.771

LEMIS (CITES-listed) vs US-reported CITES, same years

0.967

LEMIS vs global CITES 2015-2024 - the real shift

System design



Pattern-recognition techniques, mapped to the module

Supervised + unsupervised

XGBoost, Random Forest, Logistic Regression on LEMIS outcomes; peer-deviation z-score, Isolation Forest, K-Means distance without labels

Machine-learning techniques

Gradient boosting with early stopping; bagging; K-Means (k=12 by silhouette, stability ARI 0.65); isolation-based outlier detection

Hybrid / ensemble

Tree ensembles + a stacked logistic meta-learner over supervised and unsupervised signals, fitted on separate validation years

Sense-making

TreeSHAP reasons on every flag; confidence-aware triage bands; risk patterns by taxon, product, route and year

Does it work? Held-out LEMIS shipments, 2013-2014

70%

of cases caught

inspecting top 5% · random: 5%

79%

of top-1% flags real

base rate 1.6%

13%

inspected to catch 80%

random needs 80%

0.599

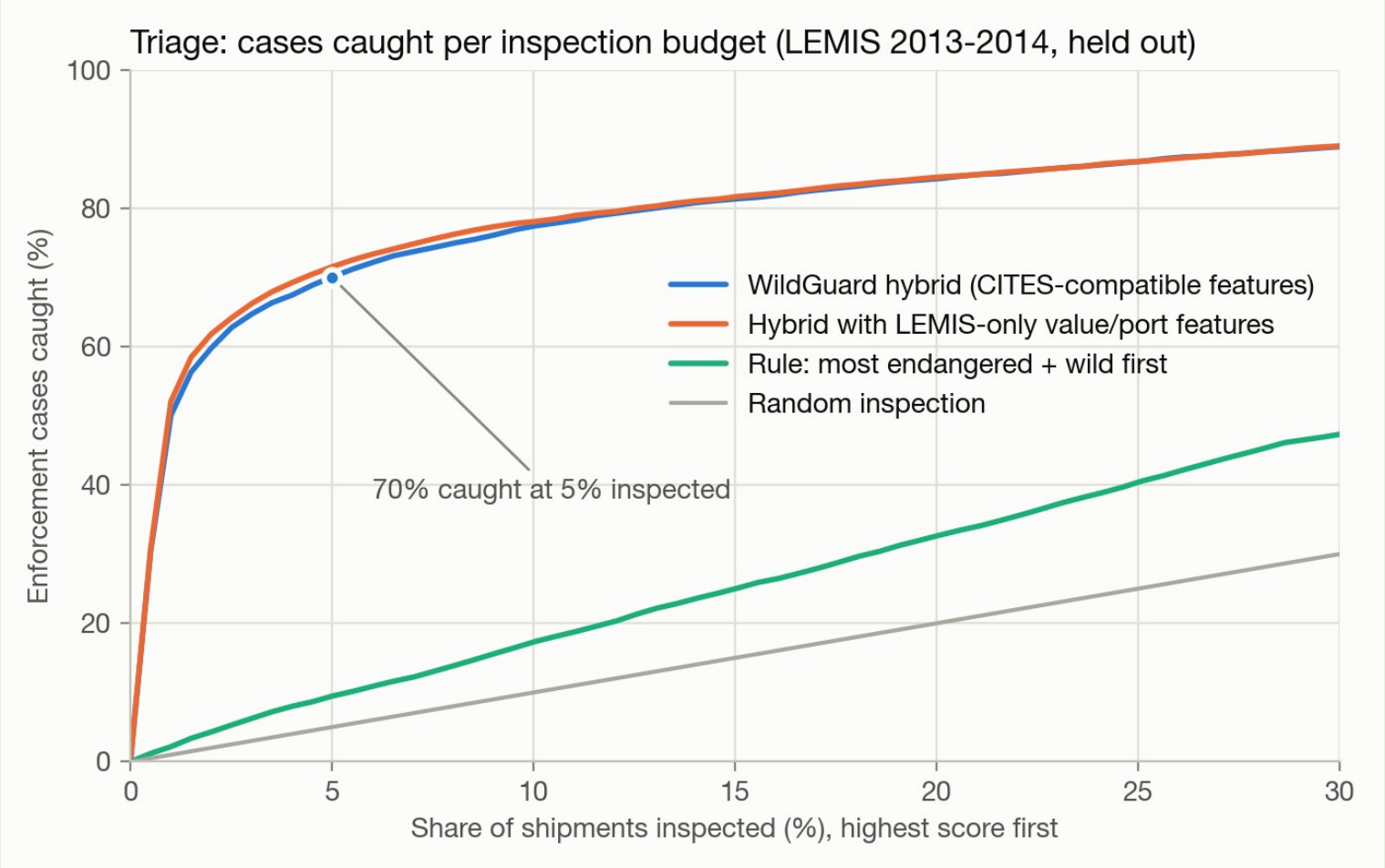
PR-AUC

random 0.016 · ROC-AUC 0.913

How it was tested

- Temporal split - train 2000-2010 (3.88M), tune 2011-2012 (813k), test 2013-2014 (822k line items)
- Enforcement case = refused, or seized / abandoned / re-exported - 1.6% of test line items, so ranking metrics (PR-AUC, recall at a budget) replace accuracy
- Deployed transfer model uses only fields CITES also has - no declared value, no US port

What an inspection budget buys



Confidence-aware bands

Auto-clear

55.5% of shipments
6.9% of enforcement cases

Review

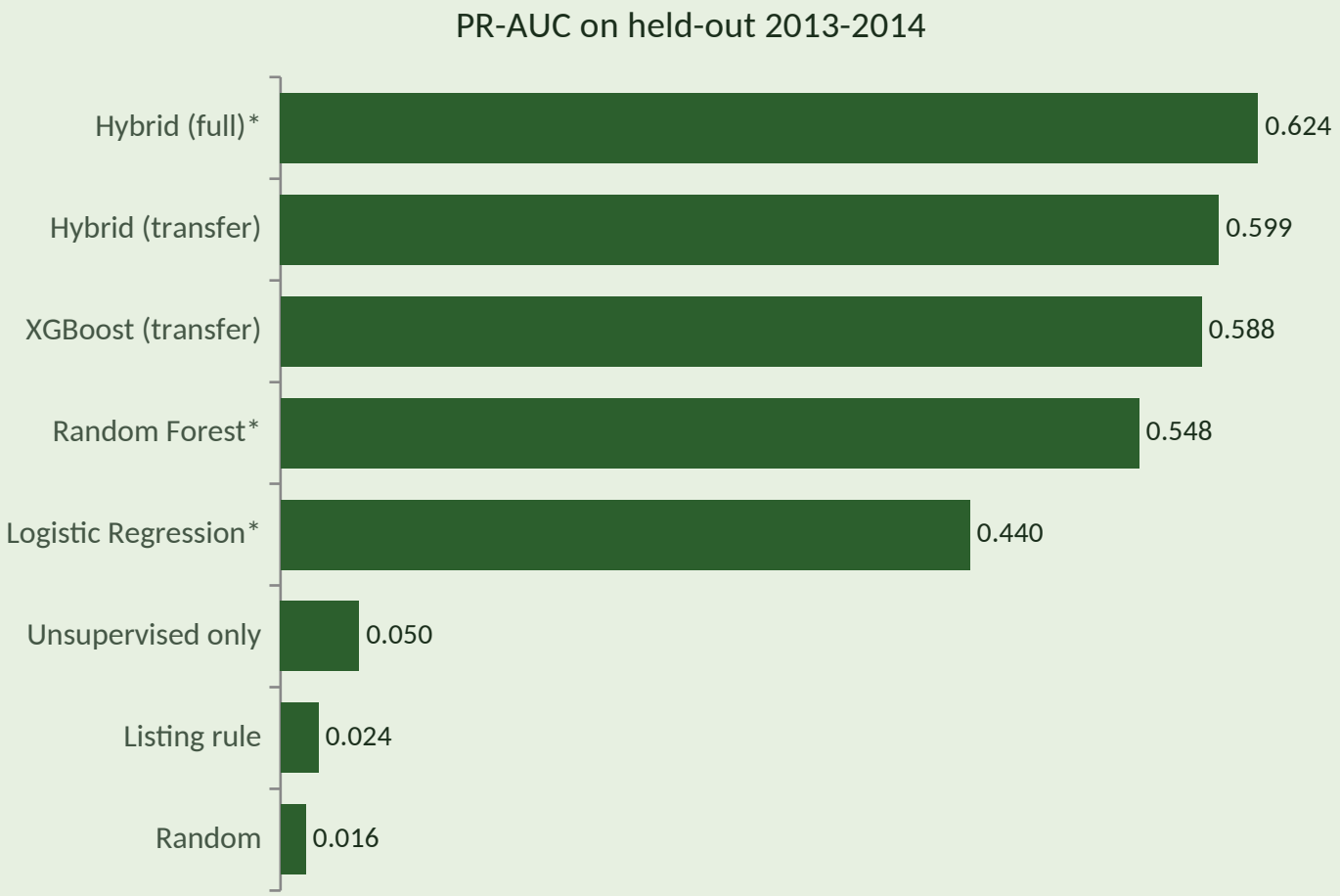
42.5% of shipments
33.5% of enforcement cases

Priority

2.0% of shipments
59.6% of enforcement cases

Thresholds fixed on 2011-2012, applied to 2013-2014

Hybrid beats either part alone - but unusual is not illegal

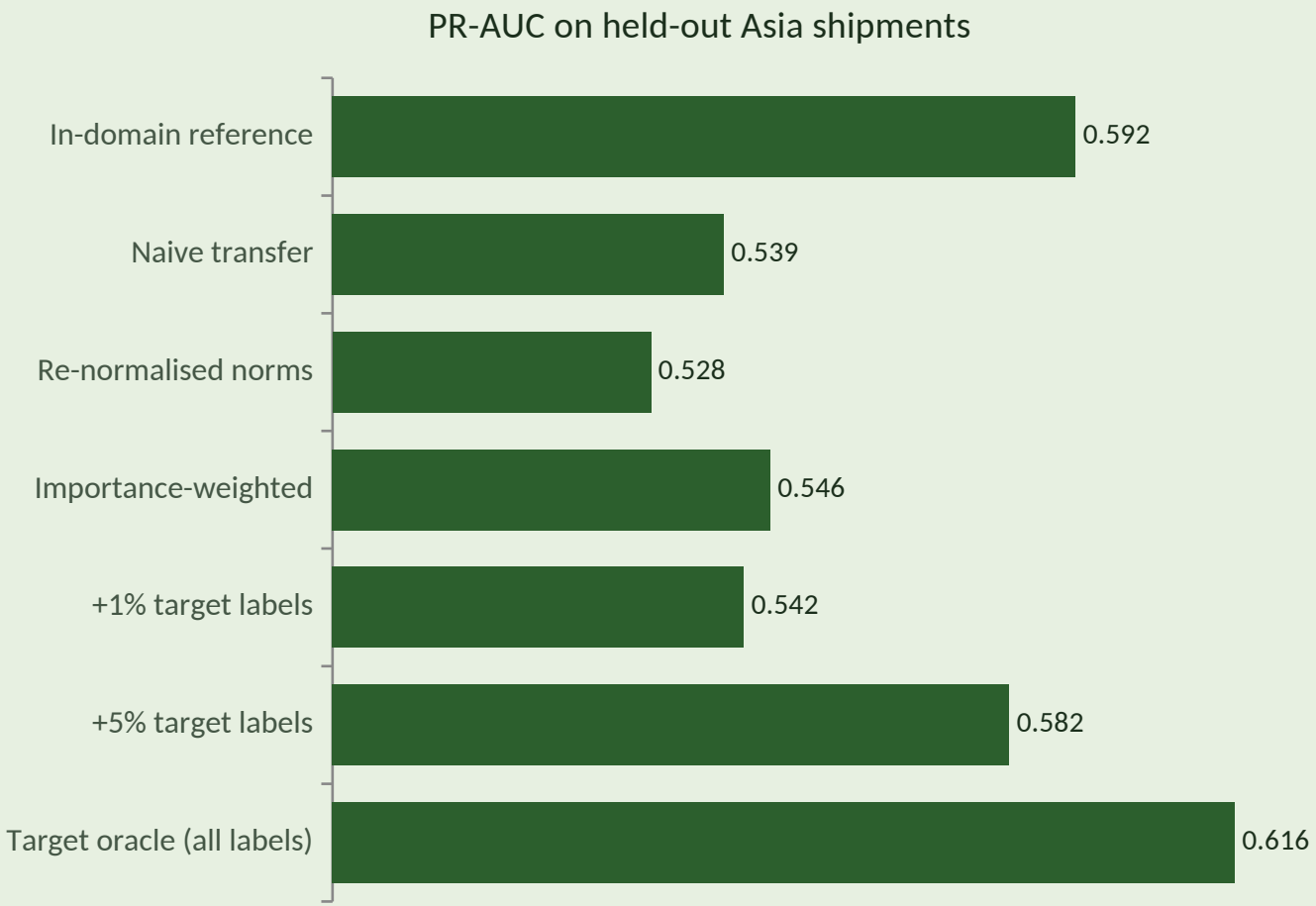


** uses LEMIS-only declared value and port*

What the combiner learned

- Weights (standardised inputs): supervised logit 0.85, Isolation Forest 1.18, peer deviation 0.36, K-Means 0.01
- Unsupervised alone: 3.1x random - legal trade is often unusual too
- Most endangered-first rule: 1.5x random - listing status alone is not the signal

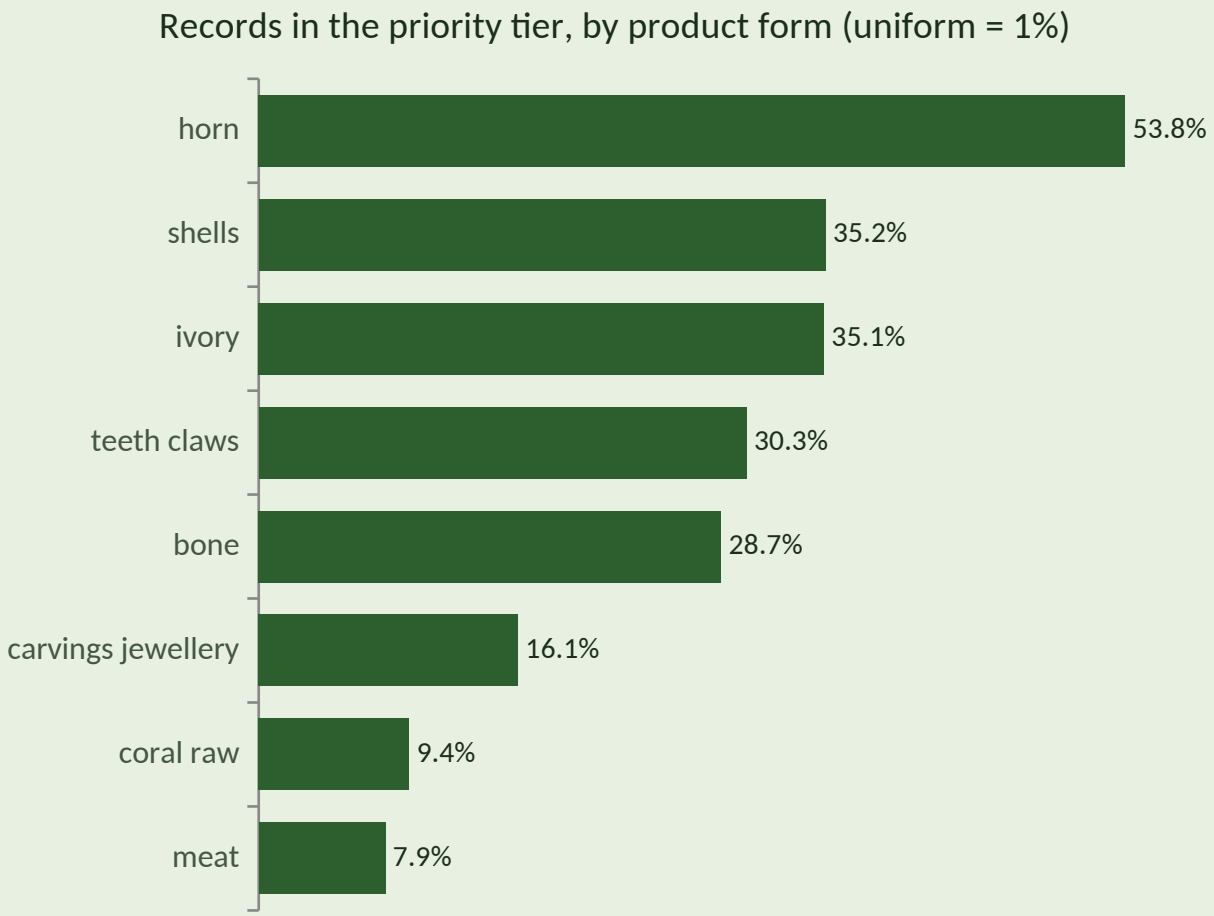
Does it transfer? A simulated shift we can actually score



Reading it

- Trained with no shipments from Asia; tested on 334k held-out Asia line items
- Re-normalising peer norms needs no labels - it is the step we apply to CITES
- A small labelled sample from the target closes much of the gap - the strongest lever for a real deployment
- Importance weighting: best here, but worse on CITES-listed LEMIS records (0.600 vs 0.630) - not deployed

What WildGuard surfaces in CITES, 2015-2024



ρ = 0.864

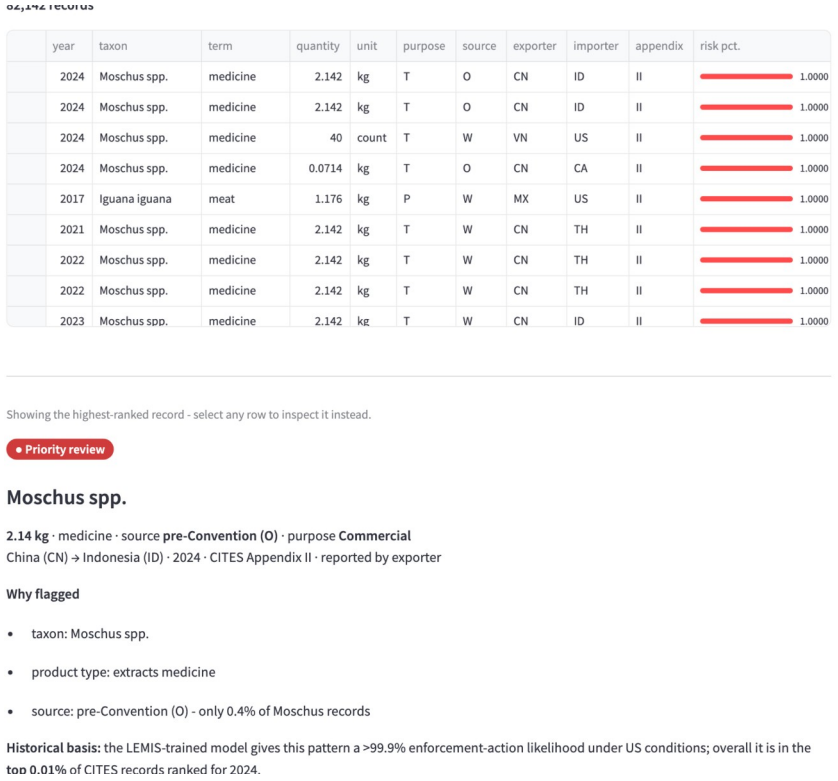
UNODC consistency (p = 0.0001, 14 species groups): flags track seizure-to-trade ratio

AUC 0.874

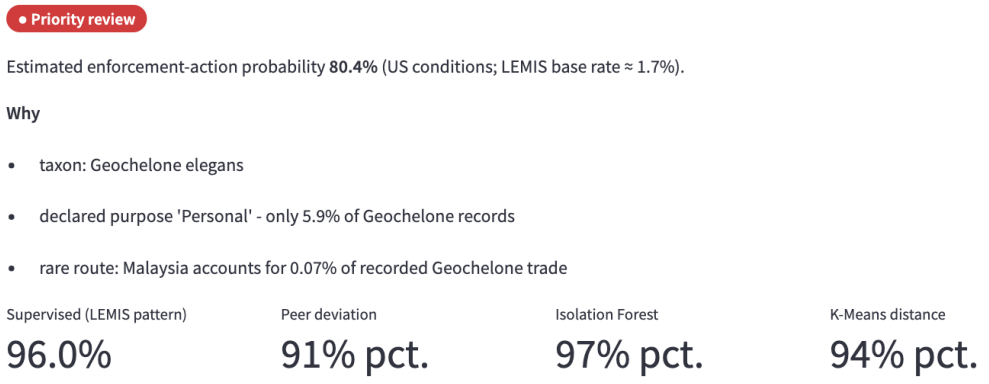
Confiscated-specimen proxy: 32,959 CITES records with source "I", ranked without the source field

Directional consistency only - no precision/recall claim is possible without CITES labels

Demo: the WildGuard review tool



Review queue + flag card



Check a shipment: the tortoise case

```
streamlit run "SYSTEM CODE/app/app.py"
```

Limitations, stated plainly

A flag is a prompt, not proof

Labels are enforcement outcomes (incl. paperwork refusals); on CITES only rankings transfer, not probabilities

Old and US-centric labels

LEMIS covers 2000-2014 and US enforcement priorities; CITES 2015-2024 is global

Personal vs commercial

Half of LEMIS cases are personal imports; on commercial shipments only, PR-AUC is 0.276 (recall@5% 52%)

Bias and feedback loops

Country features can encode where inspectors looked before - keep a human in the loop

Jurisdiction-specific labels

LEMIS refuses marine-mammal products 10x more often (US Marine Mammal Protection Act) - the model flags that trade abroad too

No CITES ground truth

Validation there is directional (UNODC, confiscated-specimen proxy), not accuracy

What WildGuard shows

- Enforcement knowledge transfers: inspecting 5% of held-out shipments catches 70% of real cases, using only CITES-compatible fields
- Hybrid > supervised > unsupervised > listing rule > random - pattern recognition beats the obvious manual heuristic by a wide margin
- Schema alignment is the enabling step: two public datasets, one shared language, verified with a domain classifier
- On CITES the ranking is directionally consistent with independent seizure statistics - a triage aid, not a verdict

Next steps

- Small labelled samples from a CITES authority (few-shot adaptation)
- LEMIS 2015+ via new FOIA releases
- Importer/exporter reporting mismatches as an extra signal
- Analyst feedback loop to retrain on reviewed flags